

Lecture 1 : An Introduction to Machine Learning

What is Machine Learning

- Arthur Samuel (1959). Machine Learning: Field of study that gives computers the ability to learn without being explicitly programmed.
- Shapire: Machine learning studies how to automatically learn to make predictions based on past observations.
- The field of machine learning tries to build and understand systems that can automatically extract information from empirical data in order to improve their performance.
- As a scientific discipline, machine learning is an interdisciplinary (and relatively young) field focusing both on theoretical foundations of systems that learn, reason and act as well as on practical applications of these systems.

What is Machine Learning

- Given a collection of examples (“training data”),
predict something about novel examples
 - The novel examples are usually *incomplete*
- Example: sorting fish
 - Fish come off a conveyor belt in a fish factory
 - Your job: figure out what kind each fish is

Machine learning draws inspiration and concepts from many scientific fields

- **Statistics:** Inference from data, probabilistic models, learning theory, ...
- **Mathematics:** Optimization theory, numerical methods, tools for theory, ...
- **Engineering:** Signal processing, system identification, robotics, control, information theory, data-mining, ...
- **Computer science:** Artificial intelligence, computer vision, information retrieval, datastructures, implementations ...
- **Economics:** decision theory, operations research, econometrics, ...
- **Psychology/Cognitive science:** Computational linguistics, learning, reinforcement learning, movement control, ...
- **Computational Neuroscience:** Neural networks, principles of neural information processing, ...

Current view of Artificial Intelligence, Machine Learning & Deep Learning

edureka!

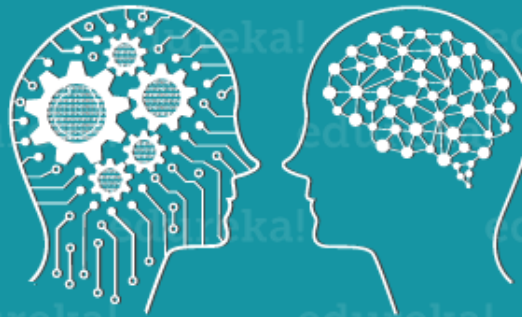
ARTIFICIAL INTELLIGENCE

Engineering of making Intelligent Machines and Programs



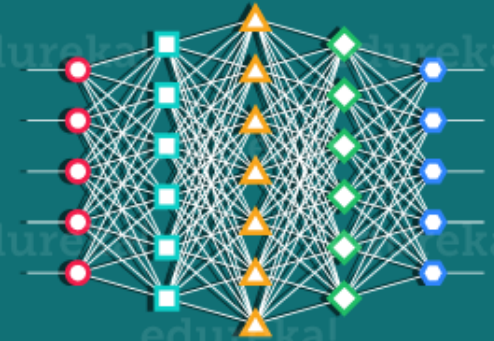
MACHINE LEARNING

Ability to learn without being explicitly programmed



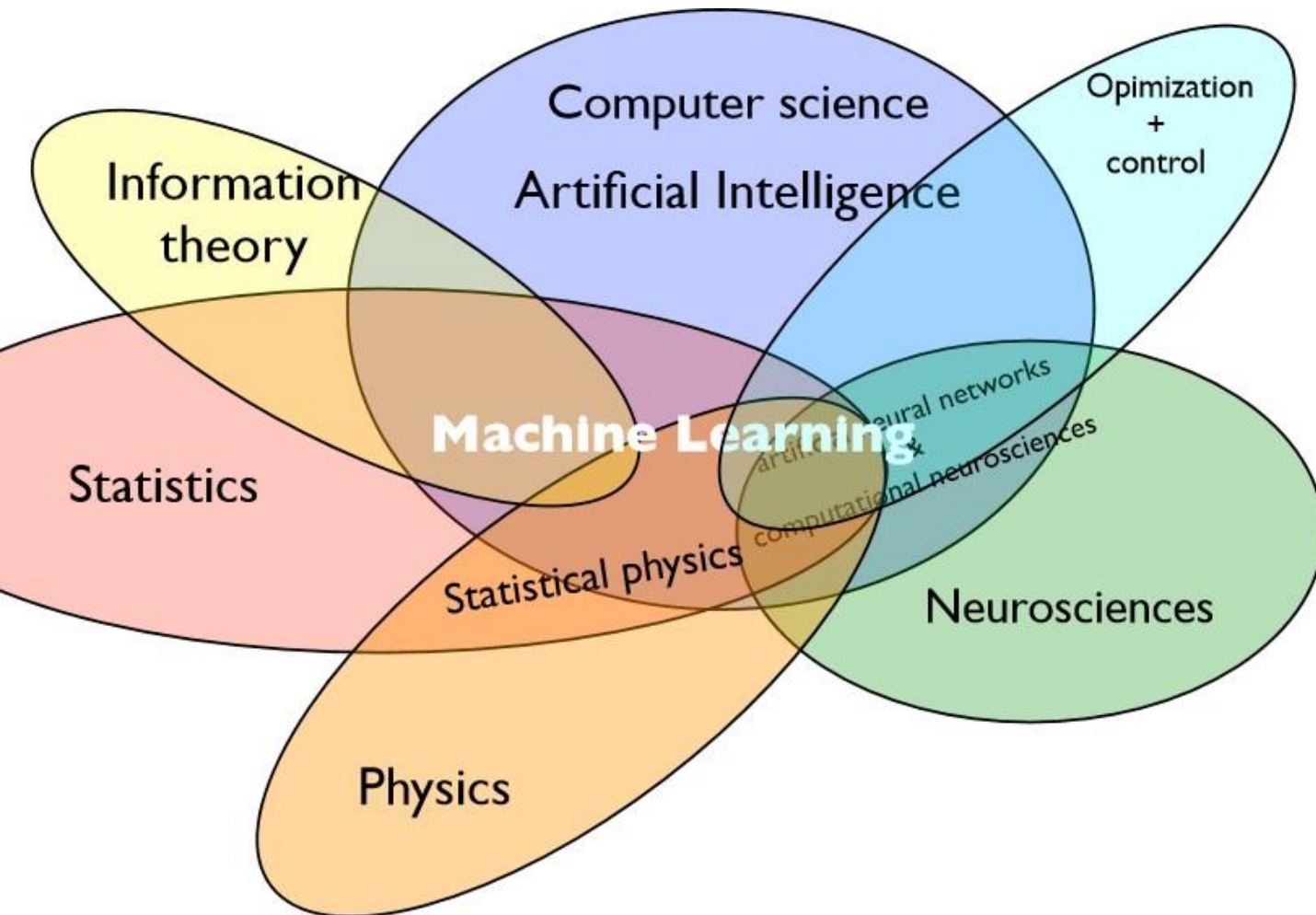
DEEP LEARNING

Learning based on Deep Neural Network



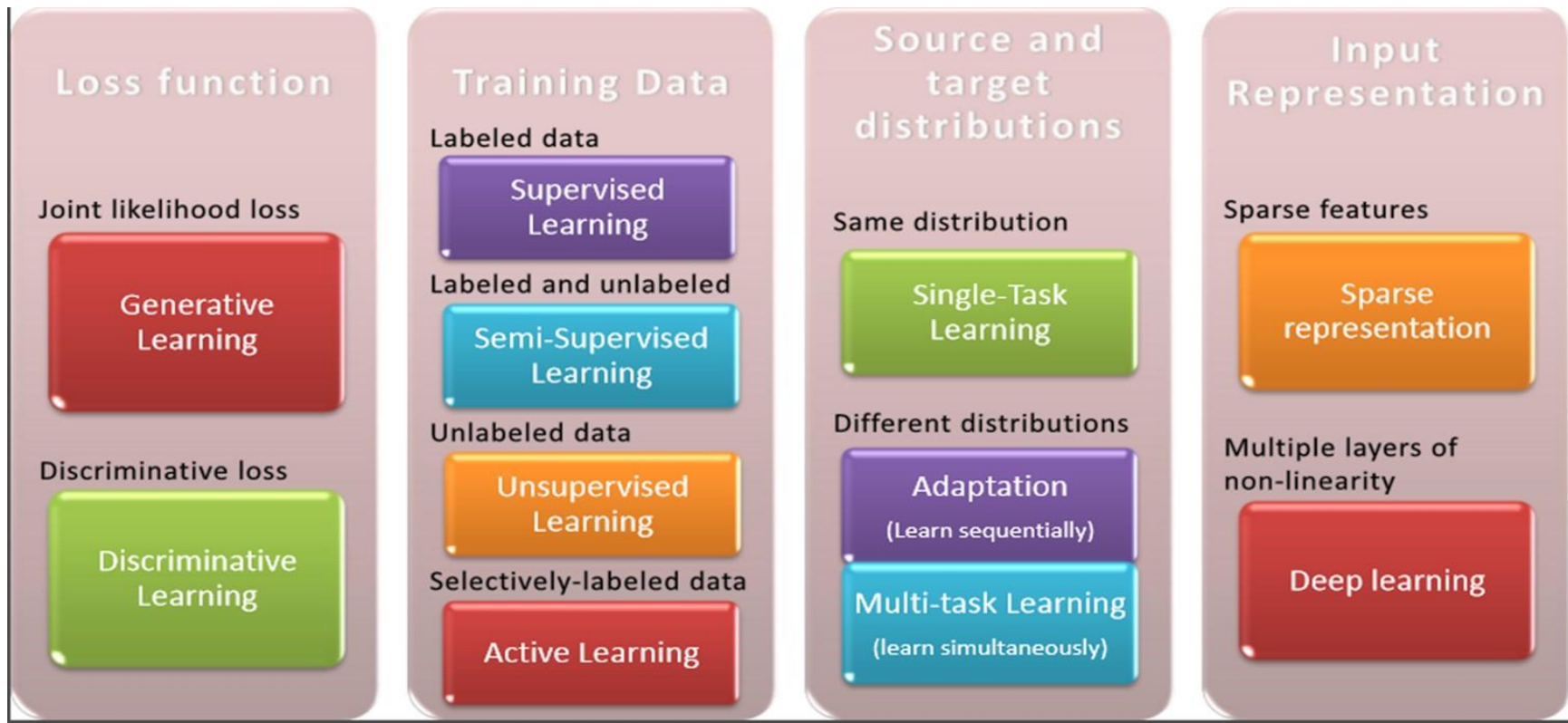
1950's 1960's 1970's 1980's 1990's 2000's 2006's 2010's 2012's 2017's

Current view of Machine Learning founding & disciplines



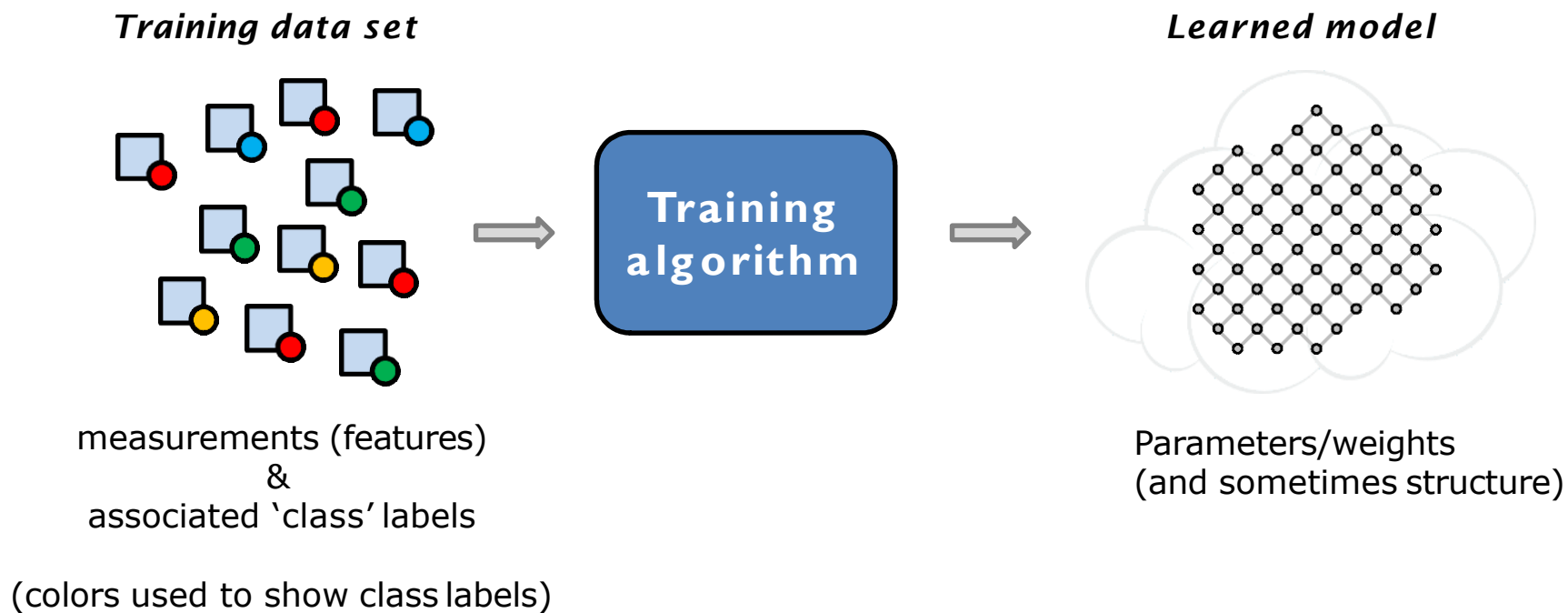
Machine Learning Paradigms : An Overview

Machine learning



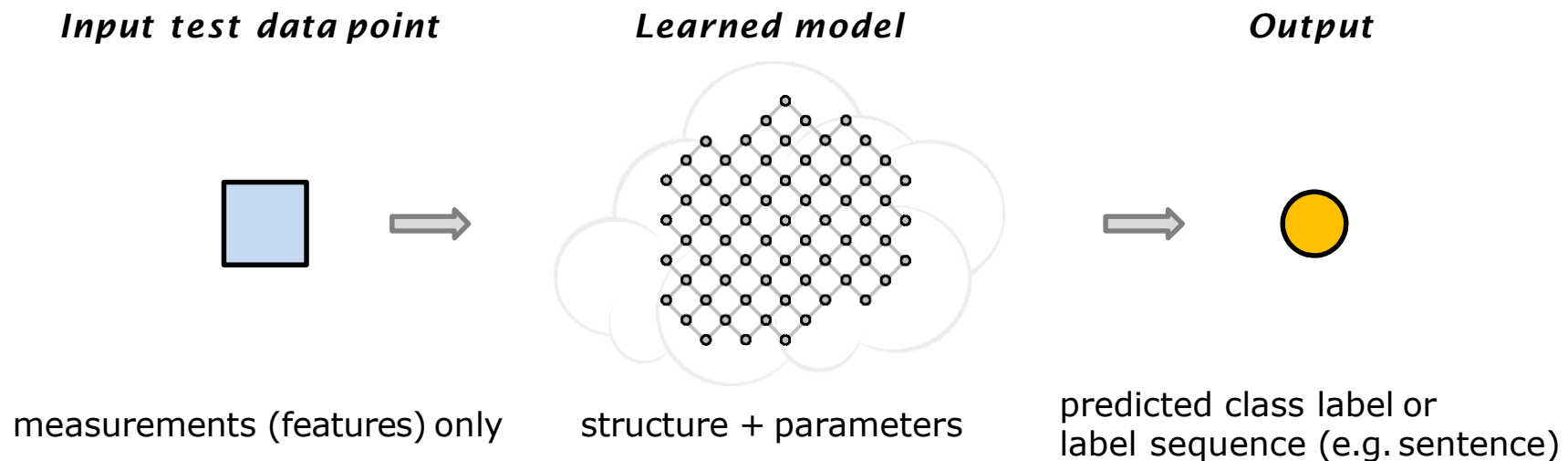
Supervised Machine Learning (classification)

Training phase (usually offline)



Supervised Machine Learning (classification)

Test phase (run time, online)



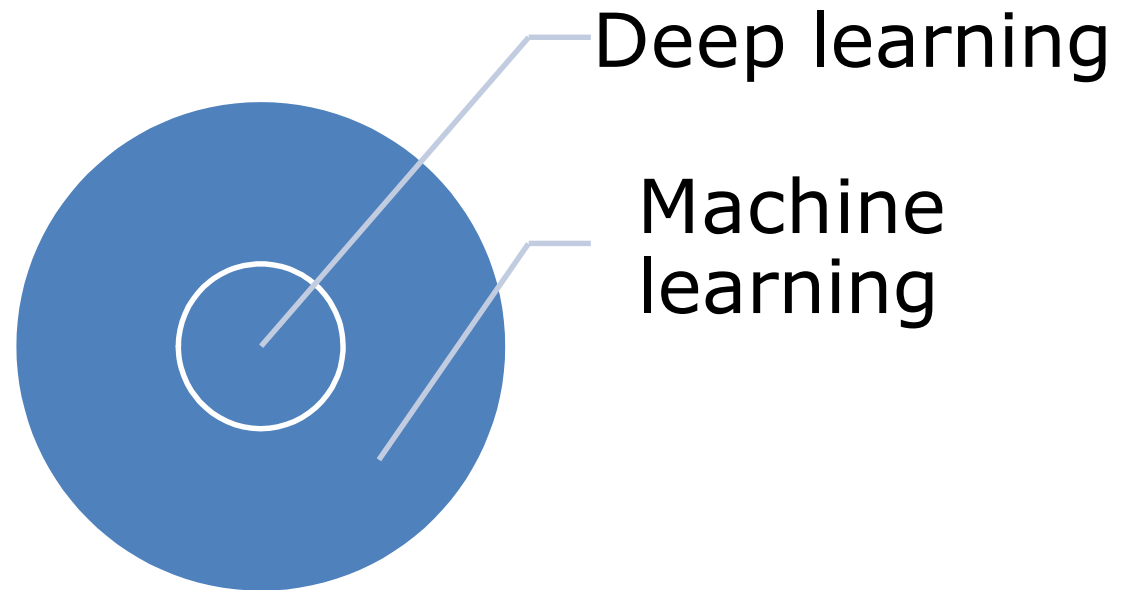
What Is Deep Learning ?



Deep learning

From Wikipedia, the free encyclopedia

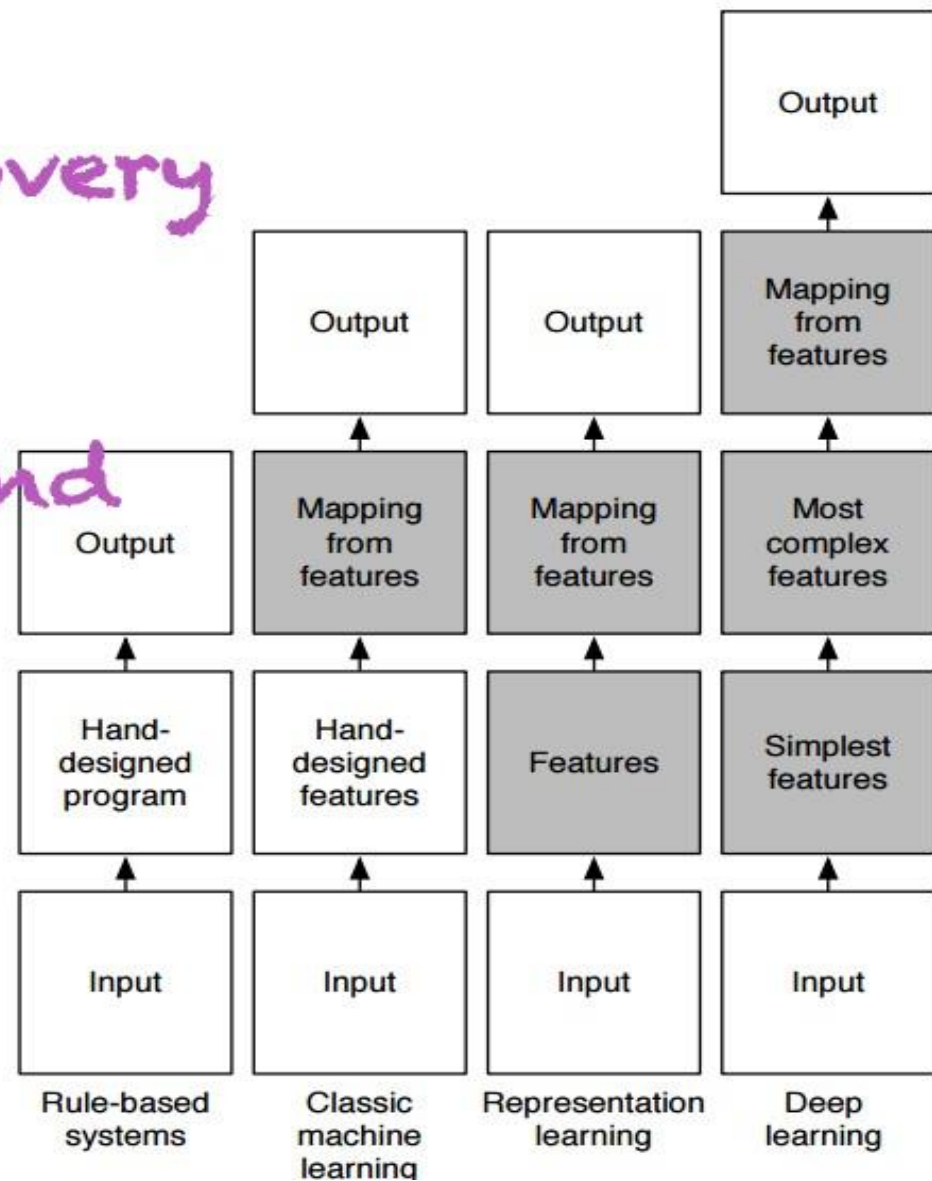
Deep learning (*deep machine learning*, or *deep structured learning*, or *hierarchical learning*, or sometimes *DL*) is a branch of [machine learning](#) based on a set of [algorithms](#) that attempt to model high-level abstractions in data by using model architectures, with complex structures or otherwise, composed of multiple [non-linear transformations](#).^{[1](p198)[2][3][4]}



Evolution of Machine Learning

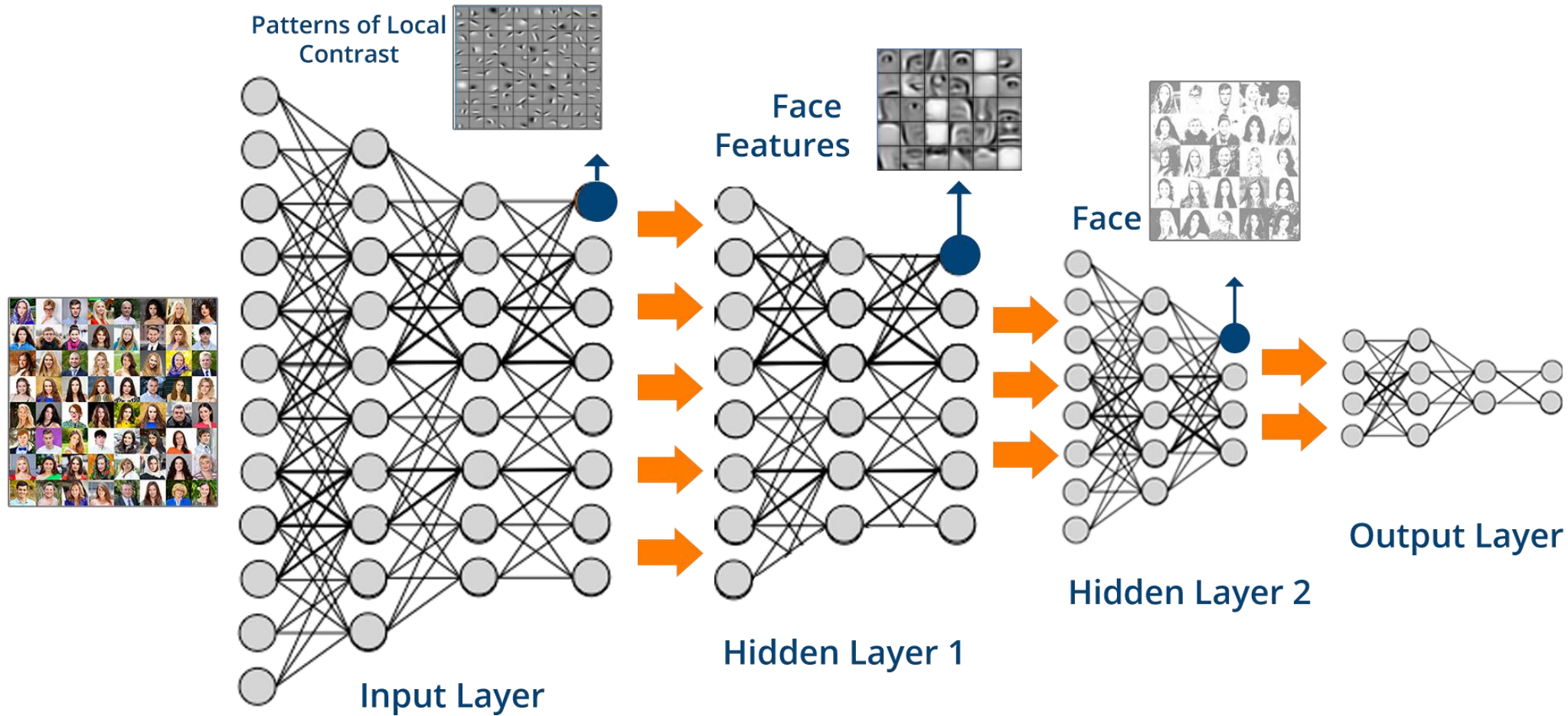
Automating
Feature Discovery

Discovering and
representing
higher-level
abstractions



(Slide from: Yoshua Bengio)

Face Recognition



SHALLOW

DEEP

Modified from
Y LeCun
MA Ranzato

Boosting

Perceptron

AE

SVM

RBM

D-AE

Neural Net

RNN

Conv. Net

DBN

DBM

GMM

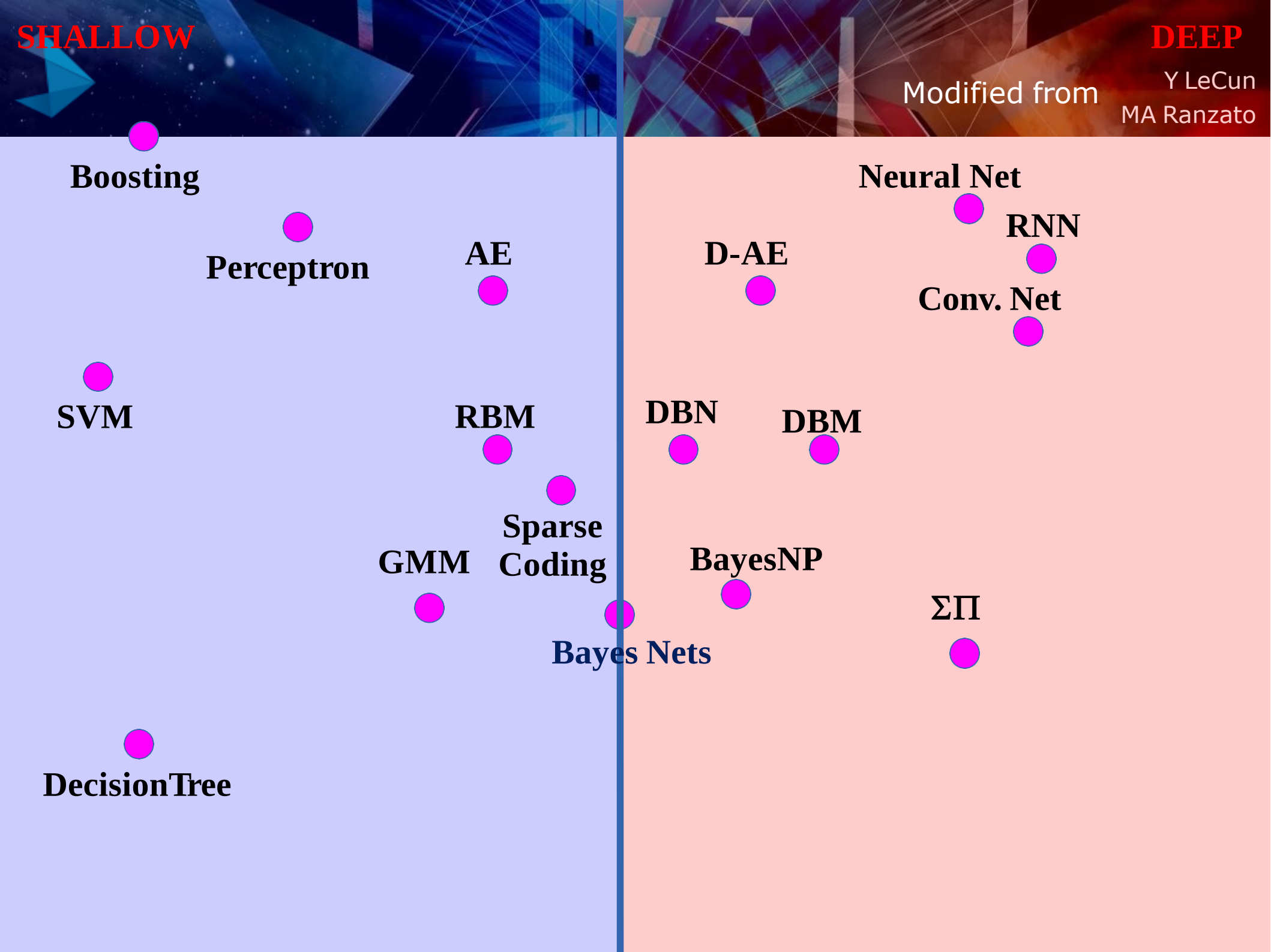
Sparse
Coding

BayesNP

$\Sigma\Pi$

DecisionTree

Bayes Nets

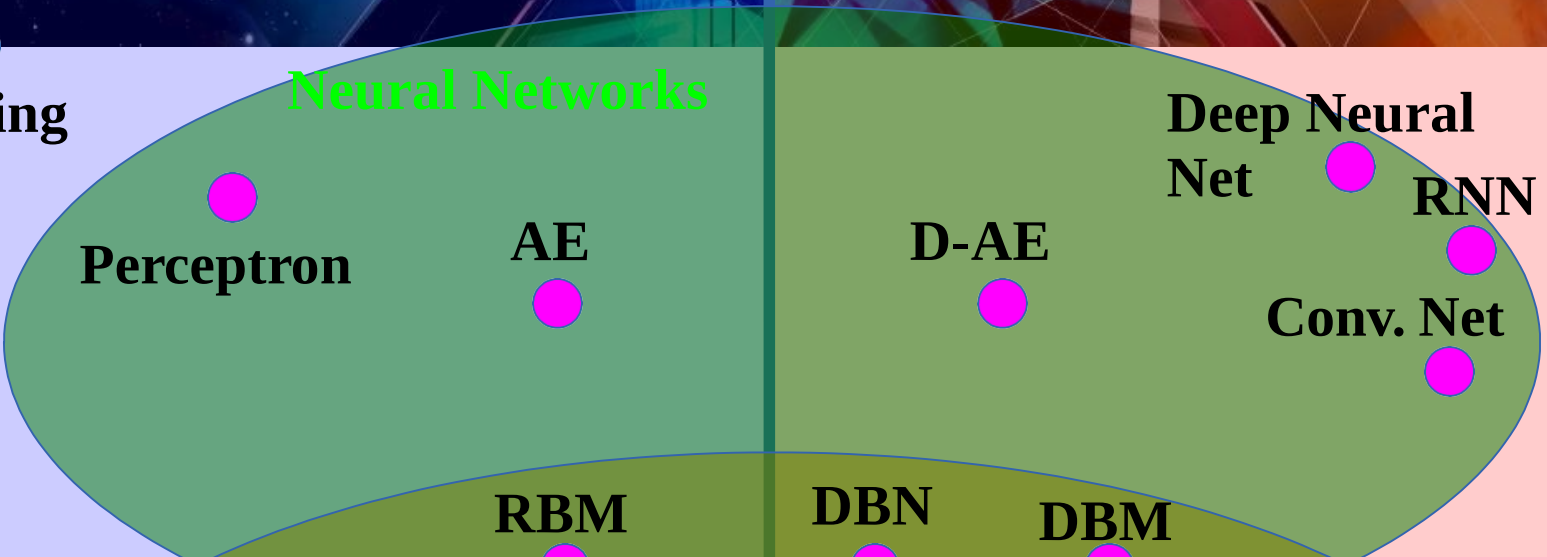


SHALLOW

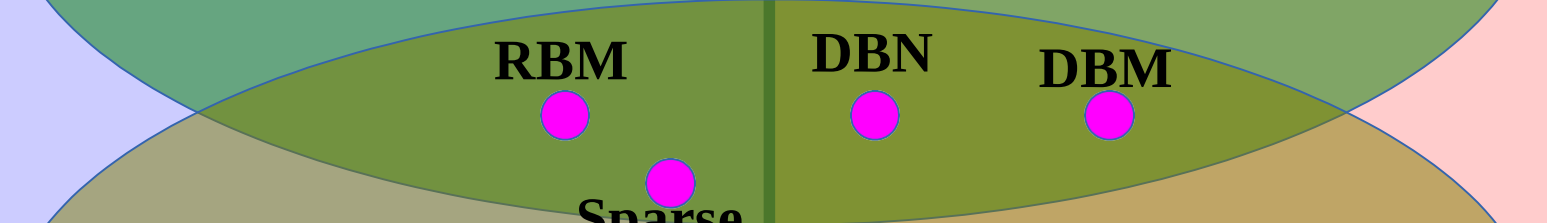
DEEP

Modified from
Y LeCun
MA Ranzato

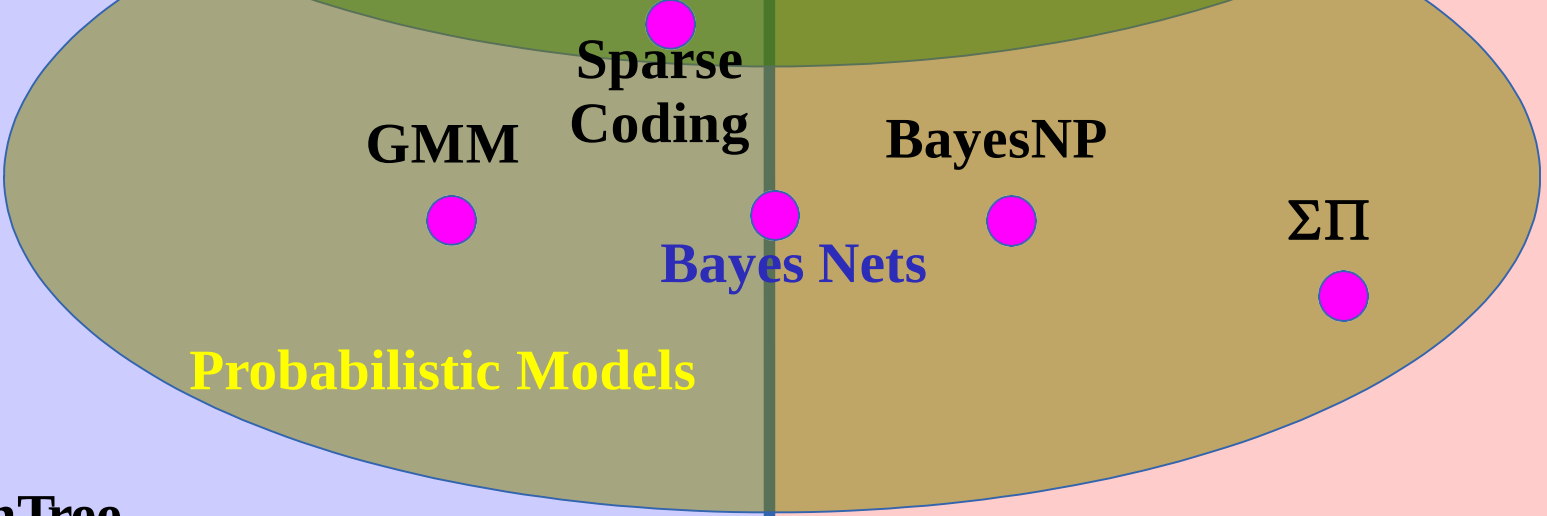
Neural Networks



RBM DBN DBM



Probabilistic Models



SVM

DecisionTree

SHALLOW

DEEP

Modified from

Y LeCun
MA Ranzato

Neural Networks

Deep Neural Net

Probabilistic Models

Unsupervised

Supervised

Boosting

Perceptron

AE

D-AE

RNN

Conv. Net

RBM

DBN

DBM

Sparse Coding

BayesNP

$\Sigma\Pi$

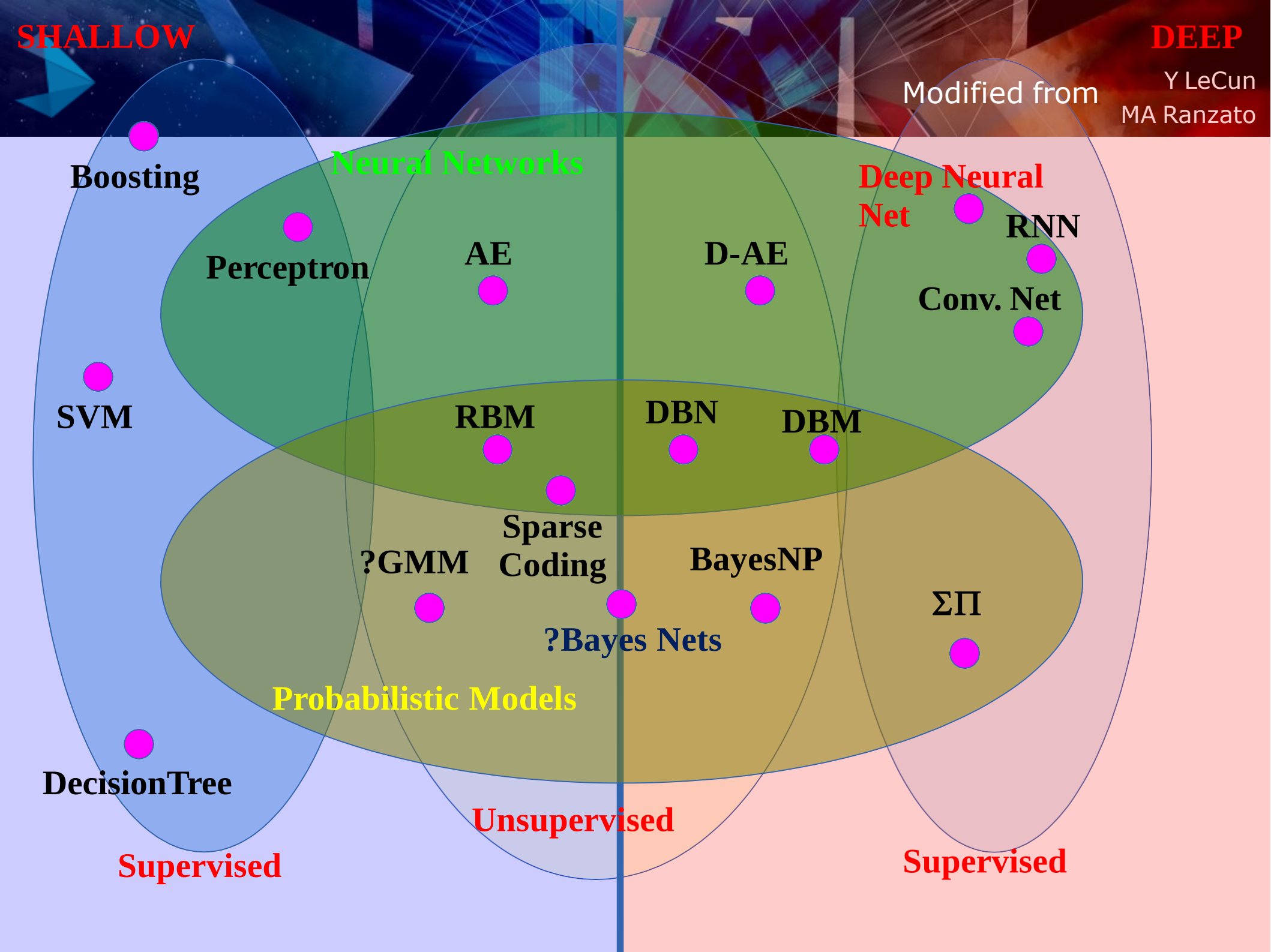
?GMM

?Bayes Nets

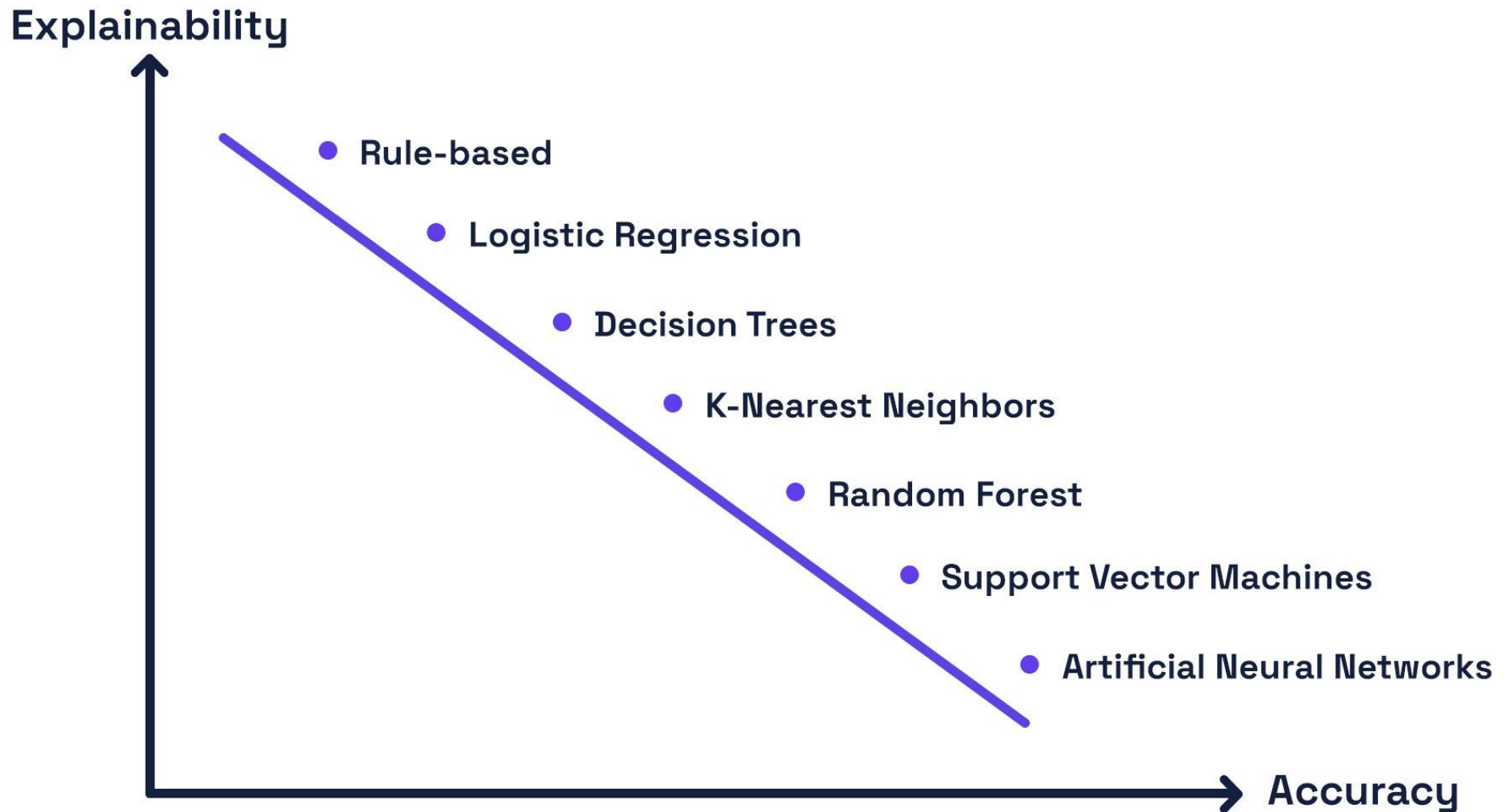
SVM

Decision Tree

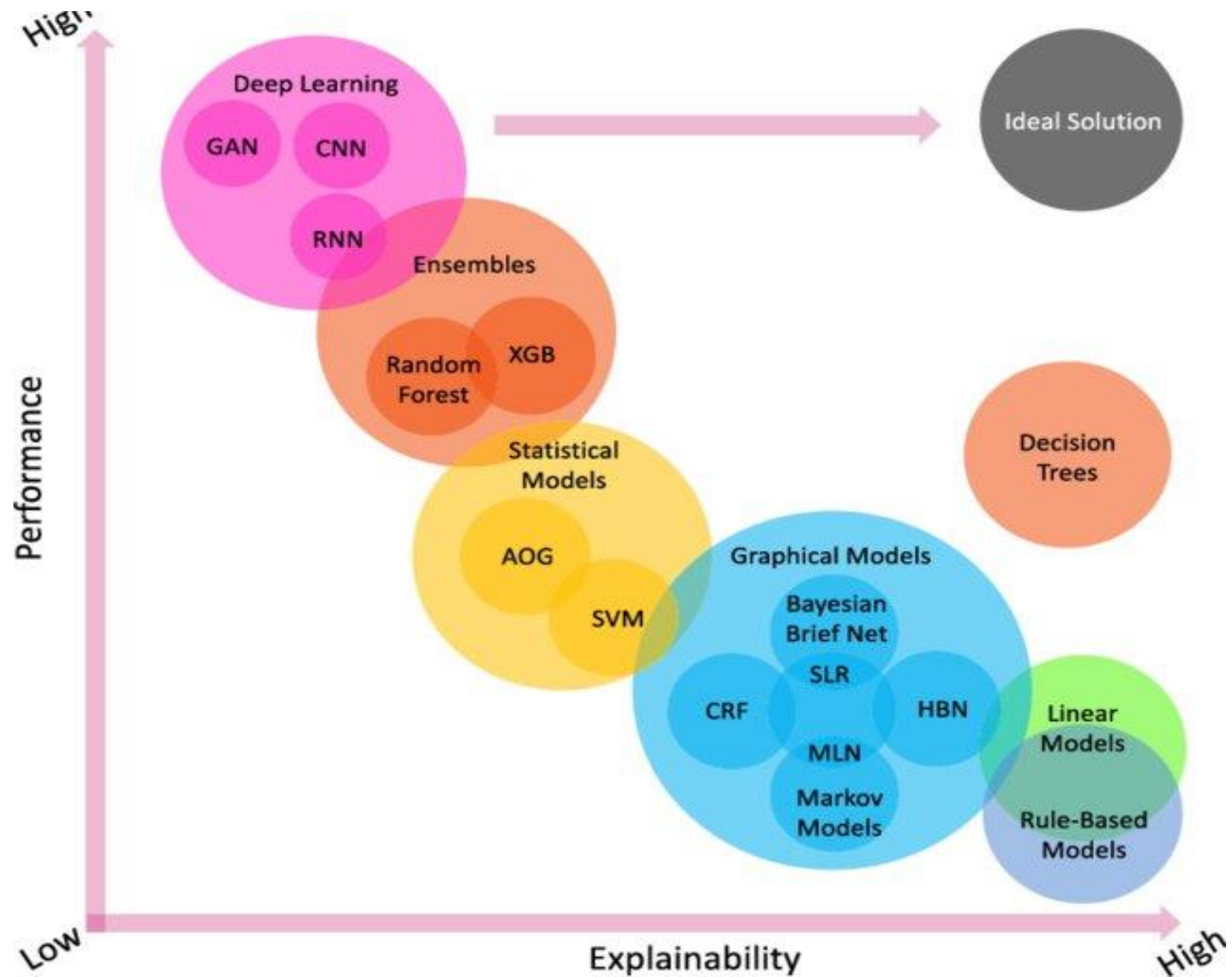
Supervised



Explainable AI (XAI)



Explainable AI (XAI)



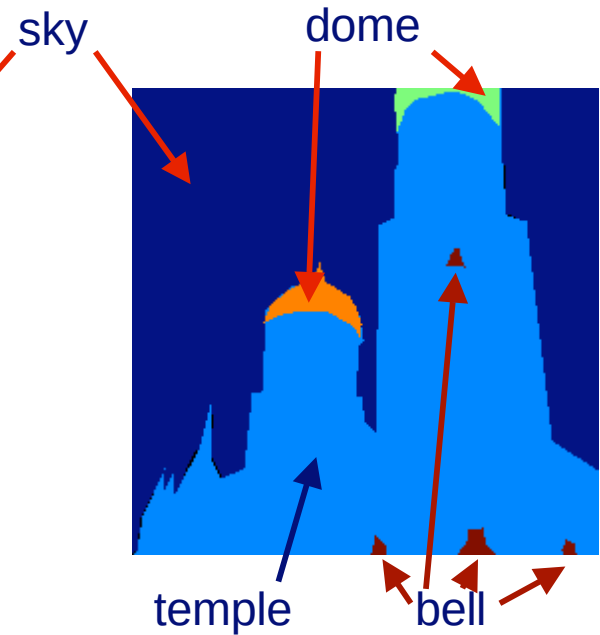
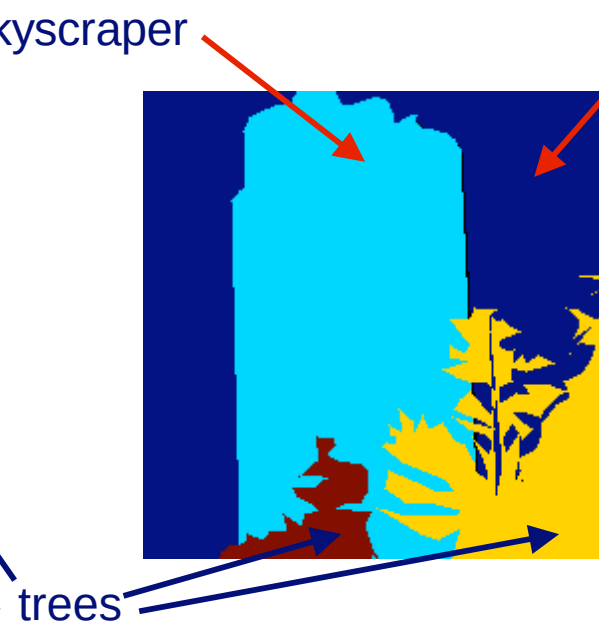
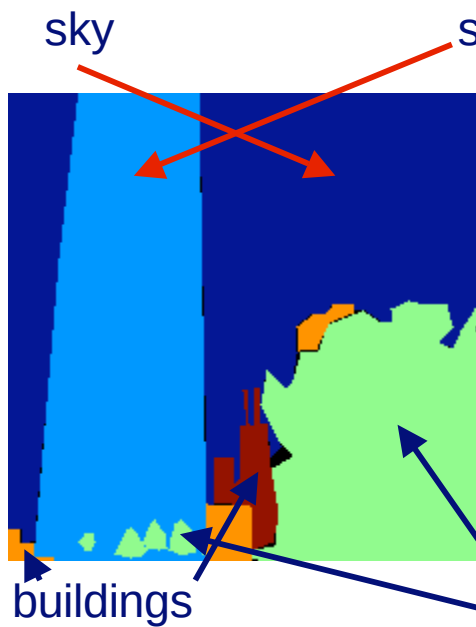
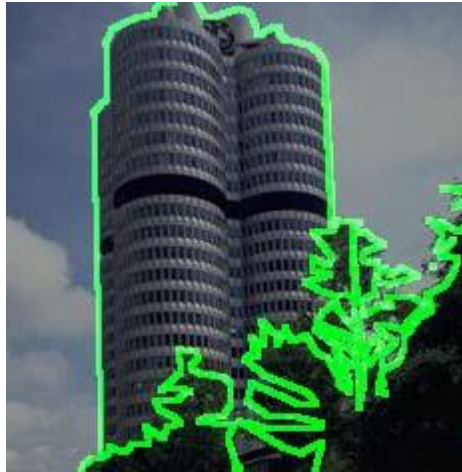
Machine Learning Problems

	<i>Supervised Learning</i>	<i>Unsupervised Learning</i>
<i>Discrete</i>	classification or categorization	clustering
<i>Continuous</i>	regression	dimensionality reduction

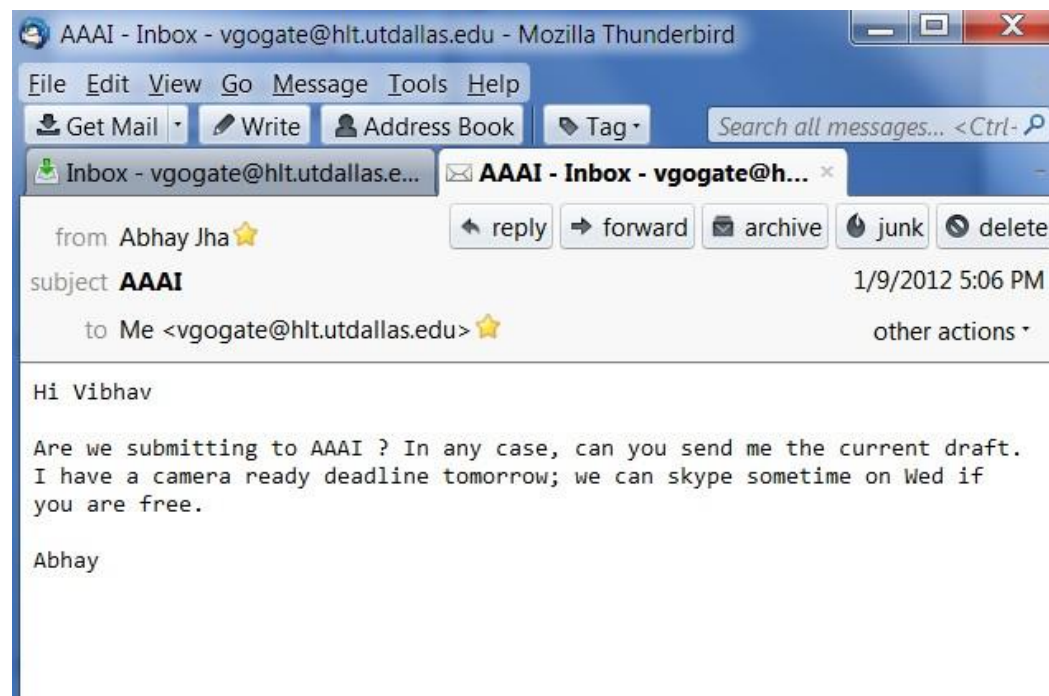
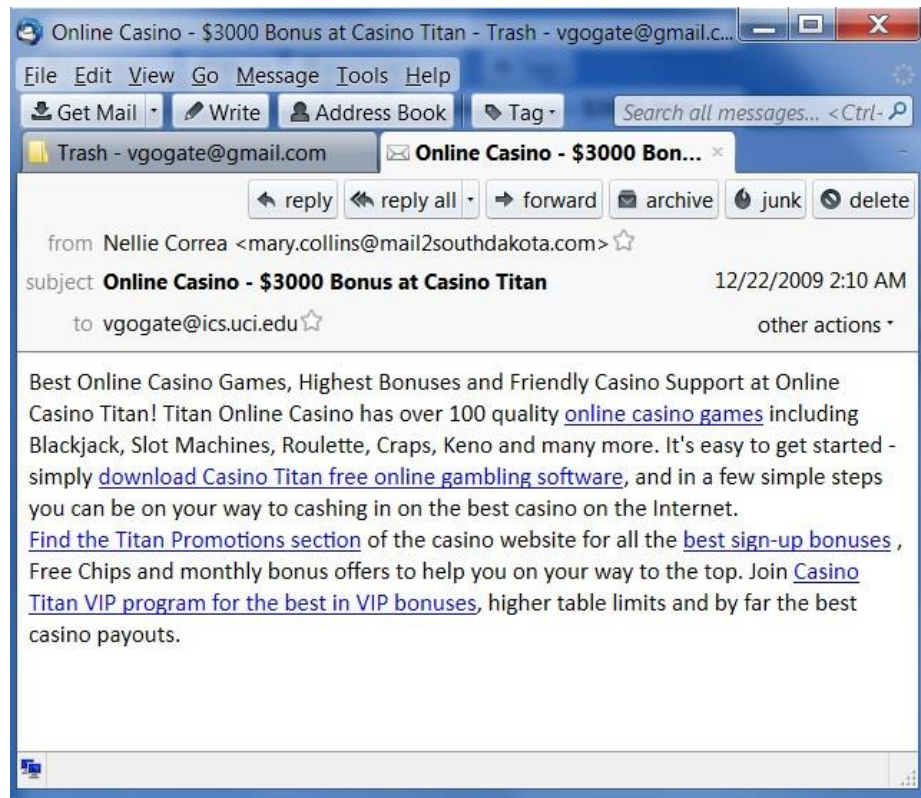
Machine Learning: Applications

Examples of what you will study in class in action!

Visual Object Recognition

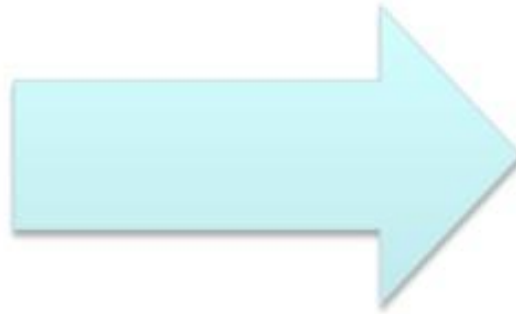


Classification Example: Spam Filtering



Classify as “Spam” or “Not Spam”

Classification Example: Weather Prediction



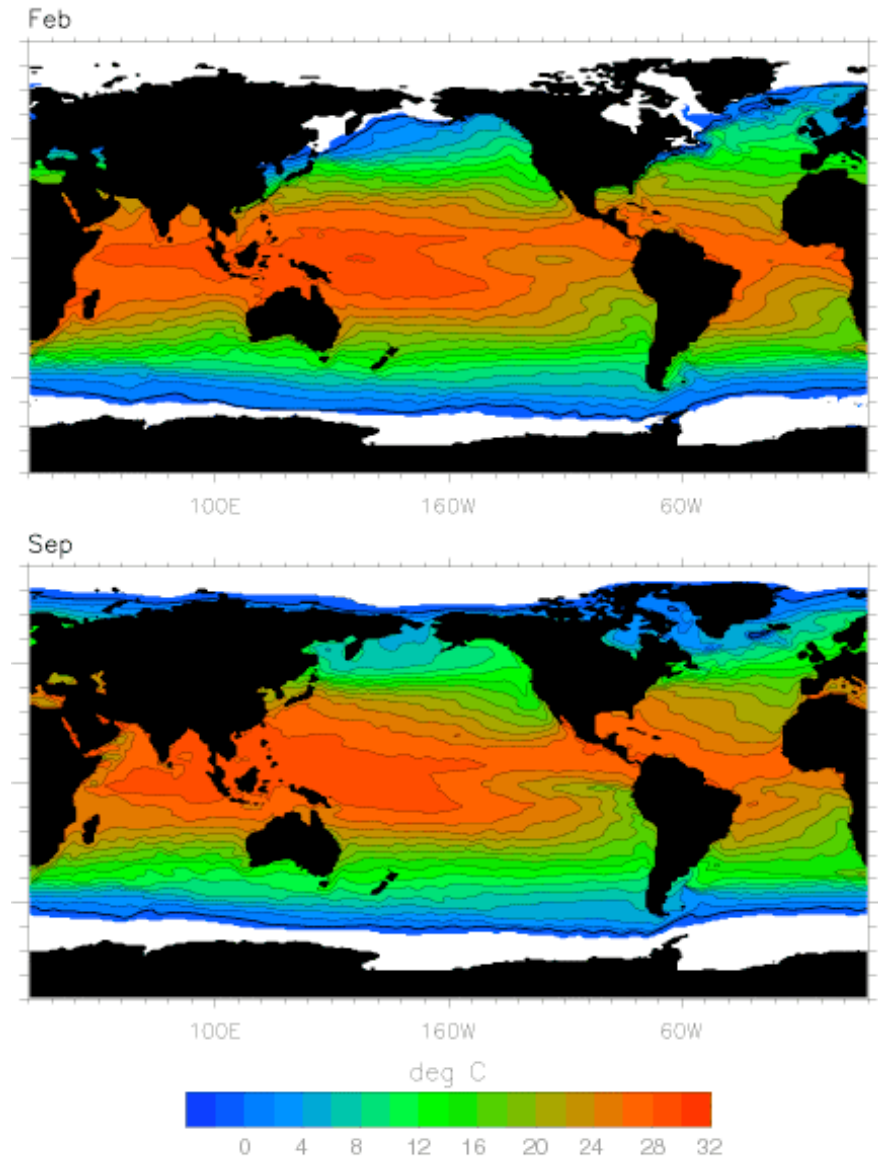
Classification: Which one of them ?
cloudy, rainy, sunny, snow



Regression :
what is Temperature Value?

Climate Modeling

- Satellites measure sea-surface temperature at sparse locations
 - Partial coverage of ocean surface
 - Sometimes obscured by clouds, weather
- Would like to infer a dense temperature field, and track its evolution



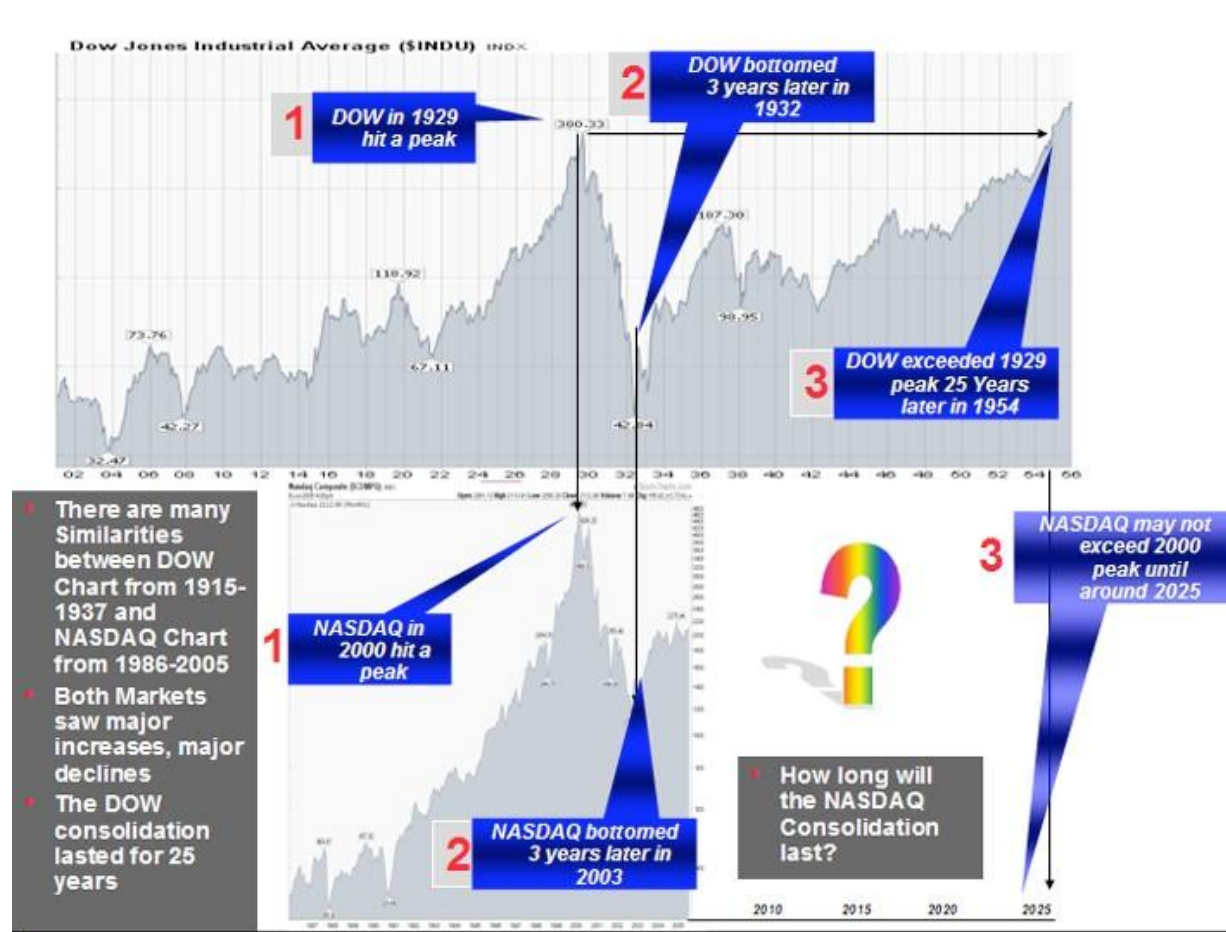
Regression example: Predicting Gold/Stock prices



Good ML can make you rich (but there is still some risk involved).

Given historical data on Gold prices, predict tomorrow's price!

Financial Forecasting



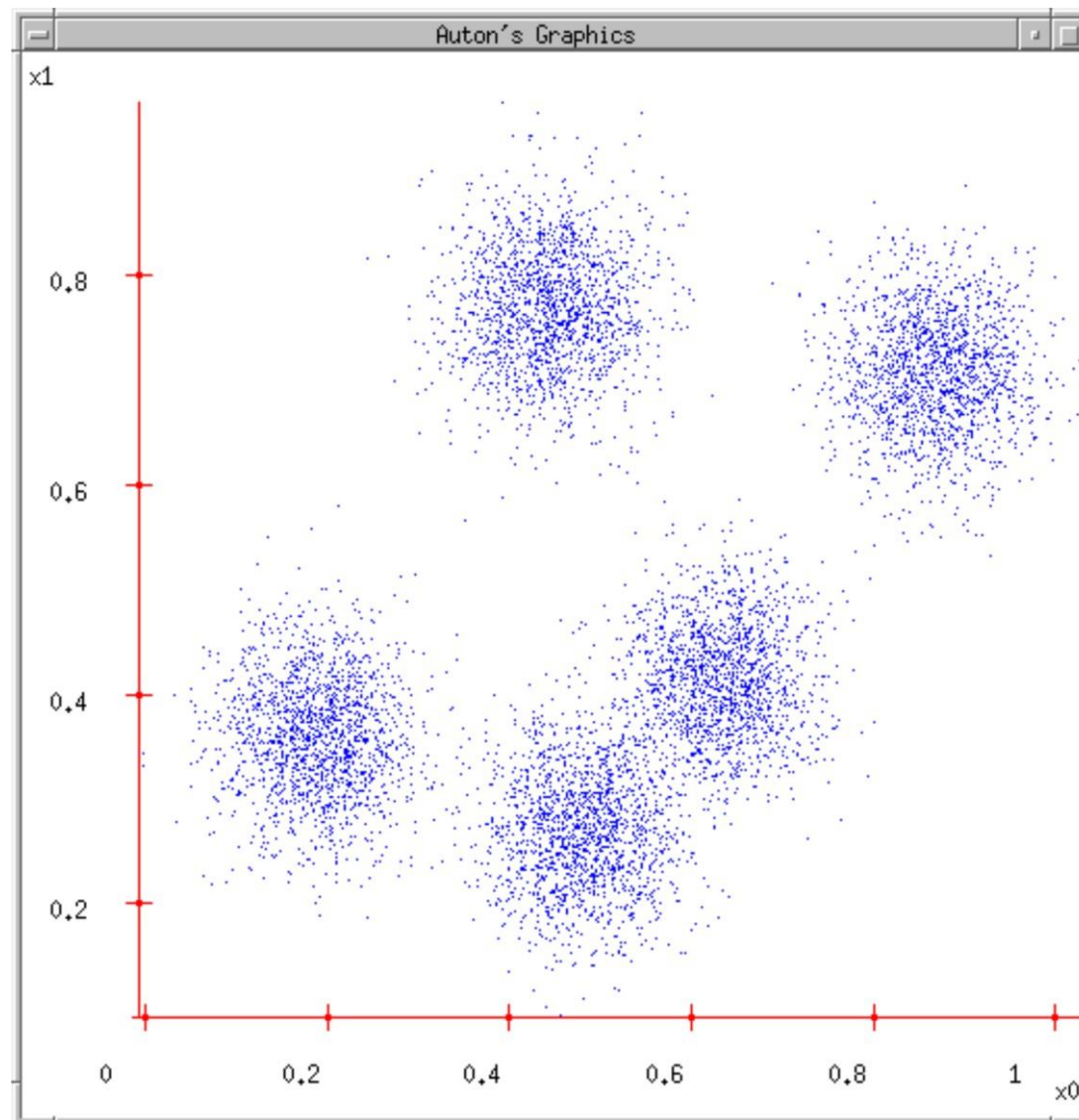
<http://www.steadfastinvestor.com/>

- Predict future market behavior from historical data, news reports, expert

Given Images: find Similar Images

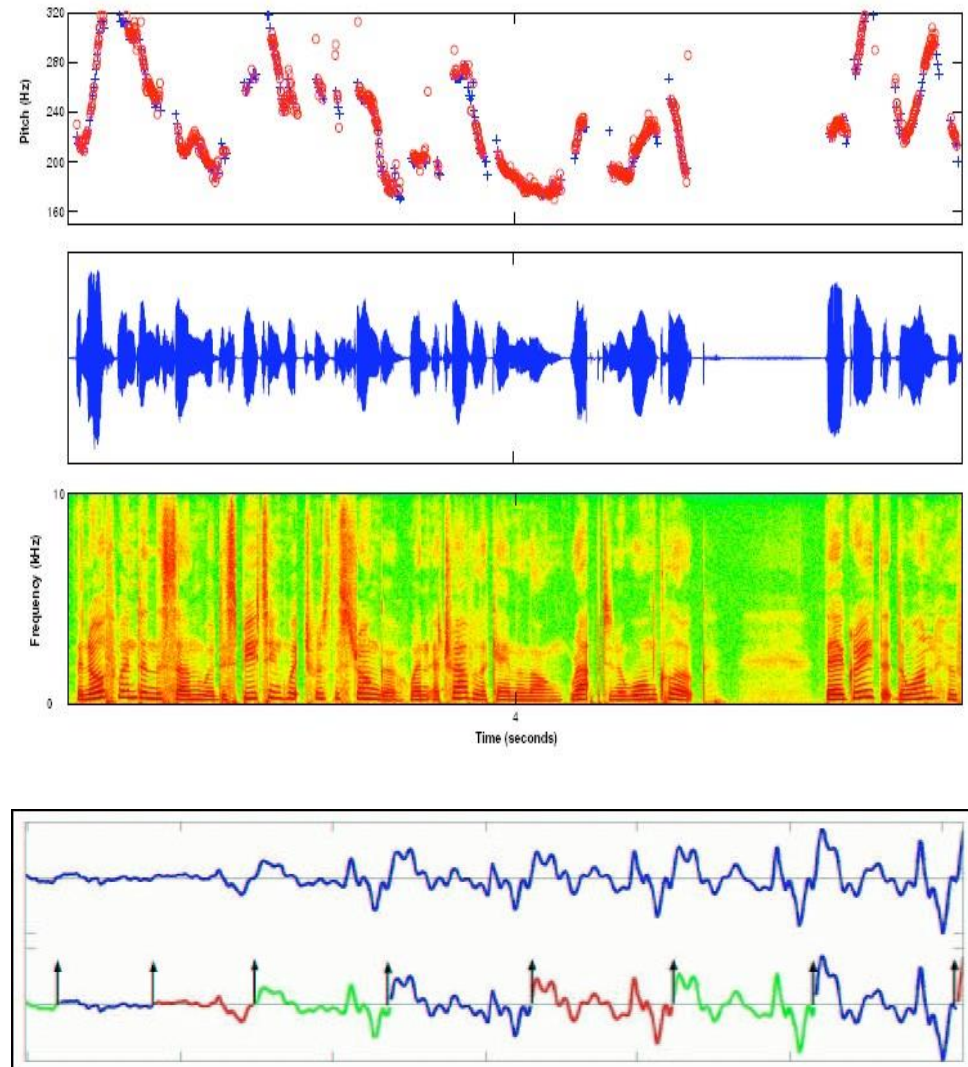
Clustering: Discover Structure in data

Group Similar Things



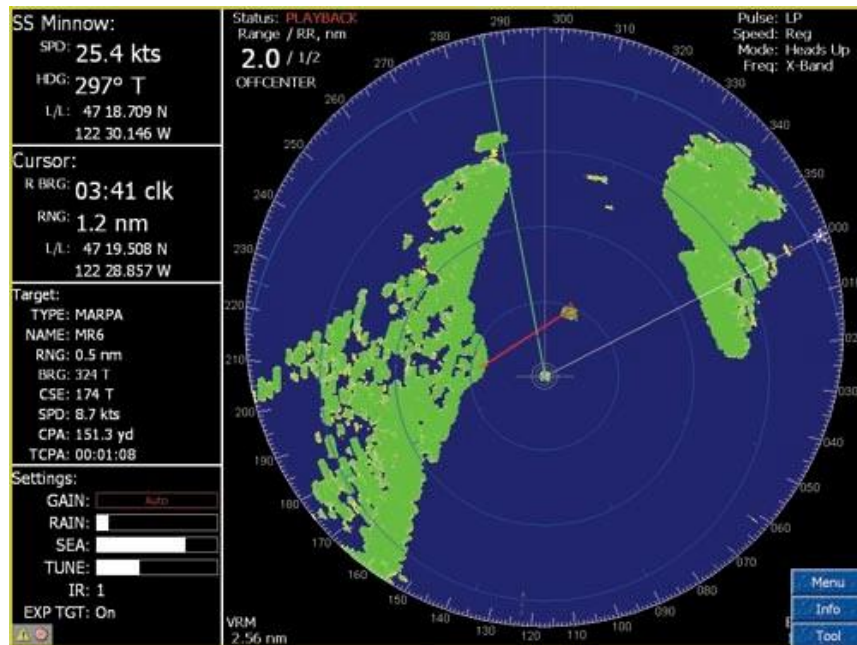
Speech Recognition

- Given an audio waveform, robustly extract & recognize any spoken words
- Statistical models can be used to
 - Provide greater robustness to noise
 - Adapt to accent of different speakers
 - Learn from training

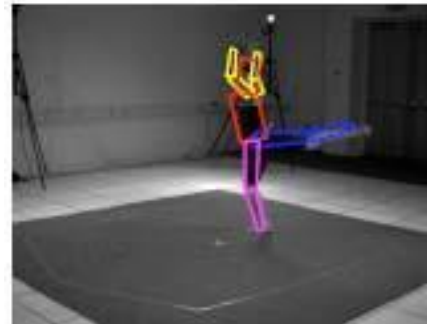
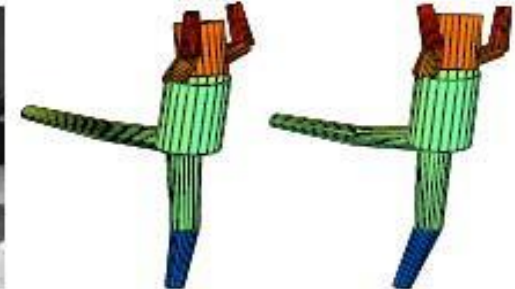
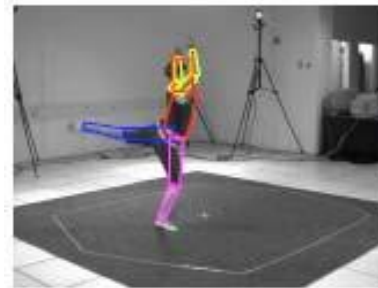


S. Roweis, 2004

Target Tracking



*Radar-based tracking
of multiple targets*



*Visual tracking of
articulated objects*

(L. Sigal et. al., 2006)

- Estimate motion of targets in 3D world from indirect, potentially noisy measurements

Machine learning has grown in leaps and bounds

- The main approach for
 - Speech Recognition
 - Robotics
 - Natural Language Processing
 - Computational Biology
 - Sensor networks
 - Computer Vision
 - Web
 - And so on

Required Basics for ML

- Prerequisites: comfort with basic
 - Programming
 - Calculus
 - Linear algebra
 - Probability